ECE 499 2026 Testing Plan Outline

ECE499 Testing Plan

- A testing plan documents the strategy that will be used to verify and ensure that a product or system meets its design specifications and other requirements

- A simple list of tests is NOT an engineering test plan

- An engineering test plan answers the following questions:
    - Why are the tests being performed?
    - What will your test results tell you?
    - How will the tests be conducted?
    - What are the resources needed?

Requirements/Specifications

Recall that you have already (in ECE498 and maybe even ECE497, detailed requirements and quantified specifications. Each requirement should be numbered as "Requirement #1", or each specification as "Specification #1", etc. This will make it easier for you to refer back to these when you discuss your testing and make it clear which requirement/specification your experiment addresses.

Testing

In the testing phase, you are expected to

(1) include the experiments performed and document experiments **in detail**

(2) explain the significance of the analysis and results **compared to the specifications.**

Given where you are at the moment, you should fill out this table **as a starting point.** If you have already designed your experiment and know what analysis you will run on the results etc., you can write out more on the Procedure/Results now which will save you time later. However, some projects may not be quite at the point where this is feasible. Your advisor will be able to guide you on their expectations.

For each of the tests that you plan to complete, fill in the table below (without the results and interpretation) to outline how the test will be conducted and the results analyzed.

Test Report - #1

Device/System/Subsystem/Component Tested: **Transportation Mode Classification Subsystem**

Objective of experiment (Engineering Specification(s) to be verified):

Verify that the on-device transportation mode classification system achieves a minimum overall classification accuracy of 90% across the 6 target transportation modes using only inertial and barometric sensor data.

Methods and Materials (or Equipment)

- Completed prototype
- Firmware implementing synchronized IMU + barometer data acquisition
- Quantized transportation mode classification model trained on the SHL dataset and self-collected data
- Laptop workstation for offline data labeling, model training, and validation
- Python-based analysis tools for accuracy and confusion matrix generation

Experimental Procedure:

1. Collect labeled IMU and barometric sensor data across each transportation mode using the fully assembled prototype.
2. Ensure each mode includes multiple trials with varied durations, speeds, and environmental conditions.
3. Preprocess raw sensor data by applying filtering, normalization, and window segmentation consistent with the training pipeline.
4. Train the classification model offline using a combined dataset (public SHL dataset and self-collected data).
5. Deploy the quantized model to the nRF52840 and enable real-time inference on-device.
6. Execute controlled validation runs where ground-truth transportation modes are known.
7. For each transportation modes, perform at least 5 independent trials.
8. Compare predicted labels against ground truth to compute classification accuracy and per-class performance metrics.

Analysis Plans:

- Compute overall classification accuracy across all test samples.
- Generate a confusion matrix to analyze per-class precision, recall, and misclassification trends.
- Evaluate class imbalance effects and identify modes with reduced performance.
- Confirm that the measured accuracy meets or exceeds the ≥90% specification threshold.

● Document any systematic classification errors.	
Results:	Interpretation:
Discussion:	
Adapted from *The Mechanical Design Process,* David G. Ullman	

Test Report - #2

Device/System/Subsystem/Component Tested: **On-Device Machine Learning Inference Subsystem**

Objective of experiment (Engineering Specification(s) to be verified):
Verify that on-device transportation mode classification inference executes with an end-to-end latency of 1 second or less per inference window while maintaining continuous sensor sampling and without reliance on external computation or cloud connectivity.

Methods and Materials (or Equipment)

- Completed Prototype
- Firmware implementing synchronized IMU + barometer data acquisition
- TensorFlow Lite for Microcontrollers runtime
- System timer and logging utilities for timestamp measurement
- USB serial interface for debug output and timing verification
- Laptop workstation for log capture and analysis

Experimental Procedure:

1. Deploy the quantized transportation mode classification model to the nRF52840.
2. Enable continuous IMU and barometer sampling.
3. Insert timing markers immediately before model inference and immediately after output generation.
4. Execute real-time inference continuously during representative movement scenarios.
5. Record inference start and end timestamps over multiple inference windows.
6. Repeat measurements at least 5 different times under varying system load conditions, including active BLE transmission.

Analysis Plans:

- Compute per-inference latency by subtracting inference start and end timestamps.
- Perform statistical analysis across repeated trials, including calculation of mean, standard deviation, minimum, and maximum inference latency.
- Verify that worst-case latency does not exceed the 1-second specification.
- Identify latency contributors such as sensor preprocessing, model execution, or BLE activity.

Results:	Interpretation:

Discussion:

Adapted from *The Mechanical Design Process,* David G. Ullman

Test Report - #3

Device/System/Subsystem/Component Tested: **Power Consumption of Li-ion Battery**

Objective of experiment (Engineering Specification(s) to be verified):
Verify that the beacon can operate continuously for at least 24 hours on a single 1200 mAh Li-ion battery while performing continuous sensor sampling, on-device inference, local data logging, and periodic Bluetooth Low Energy communication.

Methods and Materials (or Equipment)

- Completed Prototype
- Firmware implementing continuous sensing, inference, and BLE communication
- Inline current measurement tool (multimeter)
- USB power monitor
- Timestamp logging utility or stopwatch

Experimental Procedure:

1. Fully charge the 1200 mAh Li-ion battery to its nominal maximum voltage.
2. Deploy firmware configured for normal operation, including 25 Hz sensor sampling, periodic inference, and BLE transmission intervals of approximately 1 second.
3. Disconnect external power and institute standalone battery-powered operation.
4. Allow the system to run continuously under representative use conditions.
5. Monitor system status until the device shuts down due to low battery cutoff or fails to meet functional operation requirements.
6. Record total operational runtime from start to shutdown.
7. Repeat the battery drain measurement at least 5 independent times under identical operating conditions.

Analysis Plans:

- Measure total operational runtime on a single charge.
- Compare measured runtime against the 24-hour design requirement.
- Estimate average system current draw using runtime and battery capacity.
- Perform statistical analysis, such as calculating mean and standard deviation on results obtained.
- Identify high-power operational states that significantly impact battery life.

Results:	Interpretation:

Discussion:
Adapted from *The Mechanical Design Process,* David G. Ullman

Test Report - #4

Device/System/Subsystem/Component Tested: **CO_2 Estimation and Emissions Modeling Subsystem**

Objective of experiment (Engineering Specification(s) to be verified):
Verify that the system achieves a per-trip CO_2 estimation root mean square error (RMSE) of ≤ 0.20 kg when compared against ground-truth emissions estimates derived from validated distance and mode data.

Methods and Materials (or Equipment)

- Completed prototype
- Firmware implementing on-device transportation mode detection and CO_2 estimation
- Reference emissions factors for each transportation mode
- Smartphone or GPS device
- Laptop workstation for offline comparison and RMSE computation
- Python-based data analysis tools

Experimental Procedure:

1. Collect complete trip data using the beacon across all supported transportation modes.
2. For each trip, record the system's estimated CO_2 emissions based on detected mode or inferred trip duration.
3. Independently compute ground-truth CO_2 emissions using validated distance measurements and published emissions factors for each mode.
4. Align beacon-generated estimates with corresponding ground-truth values on a per-trip basis.
5. Repeat the experiment for at least 5 independent trips per transportation mode to capture variability across conditions.

Analysis Plans:

- Compute per-trip CO_2 estimation error as the difference between estimated and ground-truth values.
- Calculate the root mean square error across all evaluated trips.
- Compare the measured RMSE against the ≤ 0.20 kg specification threshold.
- Analyze error distribution across transportation modes to identify dominant contributors.

Results:	Interpretation:
Discussion:	
Adapted from *The Mechanical Design Process,* David G. Ullman	

Test Report - #5

Device/System/Subsystem/Component Tested: **User Interaction and Correction Interface**

Objective of experiment (Engineering Specification(s) to be verified):
Verify that the system requires no more than five manual user inputs or corrections per day during normal operation while maintaining accurate transportation mode detection and CO_2 estimation.

Methods and Materials (or Equipment)

- Completed prototype
- BLE-enabled smartphone or user interface dashboard
- Usage logging and interaction tracking tools
- Simulated daily transportation scenarios

Experimental Procedure:

1. Configure the system for normal autonomous operation with manual correction features enabled.
2. Simulate a full day of typical user transportation activity.
3. Record all user-initiated interactions, including corrections, confirmations, or overrides.
4. Categorize each interaction by type and time of occurrence.
5. Repeat for at least 5 independent days or usage scenarios.

Analysis Plans:

- Count total manual user inputs per day.
- Compare observed interaction count against the ≤5 inputs/day specification.
- Identify scenarios that most frequently require user intervention.
- Evaluate whether system autonomy meets usability expectations.

Results:	Interpretation:

Discussion:
Adapted from *The Mechanical Design Process,* David G. Ullman

Test Report - #6

Device/System/Subsystem/Component Tested: **System Architecture and Data Management Subsystem**

Objective of experiment (Engineering Specification(s) to be verified):
Verify that the system operates in a privacy-preserving manner by performing all transportation mode classification and CO_2 estimation locally on-device, without continuous GPS sampling and without transmitting raw sensor or location data to external servers.

Methods and Materials (or Equipment)

- Adafruit Feather nRF52840 Express microcontroller
- Adafruit BNO055 IMU (accelerometer, gyroscope)
- Adafruit BMP390 barometric pressure sensor
- 1200 mAh Li-ion battery
- Firmware implementing local data storage and optional BLE-based summary transmission
- BLE-enabled host device for verification of transmitted data contents
- Serial logging interface for debugging and inspection

Experimental Procedure:

1. Deploy firmware configured for normal operation with GPS functionality disabled or limited to manual calibration only.
2. Operate the device across representative transportation scenarios.
3. Monitor system behavior to confirm that all sensor data acquisition, inference, and CO_2 estimation occur locally on the microcontroller.
4. Capture BLE transmissions and inspect transmitted payloads.
5. Verify that only aggregated summaries or classification outputs are transmitted, with no raw sensor data or continuous location information included.

Analysis Plans:

- Confirm absence of continuous GPS sampling during normal operation.
- Verify that no raw IMU, barometric, or location data is transmitted over BLE.
- Confirm that all classification and emission estimation occur on-device.
- Validate compliance with the system's stated privacy and data-minimization goals.

Results:	Interpretation:
Discussion:	
Adapted from *The Mechanical Design Process,* David G. Ullman	